

GOLD IN
QUARTZGOLD
NUGGET**GOLD**

Cubic • H 2½–3 • SG 19.3

Prized for its colour and malleability, gold forms in hydrothermal veins and often weathers out to be found as nuggets in river sands.

**IRON**

Cubic • H 4½ • SG 7.3–7.87

Most native iron occurs at Earth's core, and near the surface it readily combines with other elements.

**BISMUTH**Trigonal
H 2–2½ • SG 9.7–9.83

Native bismuth is relatively rare. Hardly ever found as distinct crystals, it more often has a granular or branching form.

mercury
globules in
rock cavities**MERCURY**Trigonal
H Liquid • SG 14.38

This is the only metal that is liquid at normal temperatures. In liquid form, mercury appears as silvery globules.

**SILVER**

Cubic • H 2½–3 • SG 10.5

Widely distributed but rarely found in abundance, native silver occurs mainly as twisted wires, scales, and branching masses.

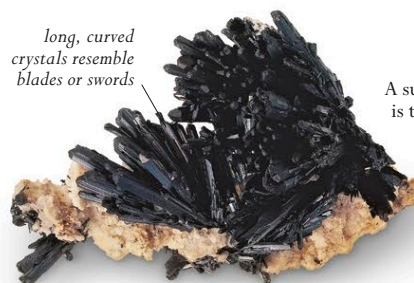
SULFIDES

The sulfides are a large group of minerals, in which sulfur is combined with one or more metals. Many sulfides have a high specific gravity and a metallic lustre. They often form as excellent crystals. Sulfides occur in many geological situations, but frequently in hydrothermal veins. The group includes the majority of the economically important metal ore minerals.

**COBALTITE**

Orthorhombic • H 5½ • SG 6.33

Cobaltite is an uncommon sulfide of arsenic and cobalt. It is an important cobalt ore in, for example, Sweden and Norway.

long, curved
crystals resemble
blades or swords**STIBNITE**

Orthorhombic • H 2 • SG 4.63–4.66

A sulfide of antimony, this dark grey mineral is the main ore of antimony. Large deposits occur in China, Japan, and W. USA.

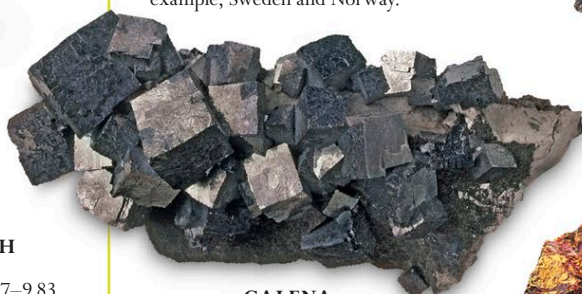
BORNITE
CRYSTALS**BORNITE**

Orthorhombic • H 3 • SG 5.08

This copper iron sulfide is a coppery red colour, tarnishing to shimmering purple and blue. It is an important copper ore.

indistinguishable
crystals form
large masses

MASSIVE BORNITE

**GALENA**

Cubic • H 2½ • SG 7.58

Lead sulfide is one of the more abundant and widely distributed sulfide minerals. It is mined extensively as lead ore.

**GREENOCKITE**Hexagonal
H 3–3½ • SG 4.82

Named after Lord Greenock, on whose Scottish property it was discovered in 1840, this rare cadmium sulfide can be yellow, red, or orange.

USUAL,
CRYSTALLINE
FORM OF
SPHALERITEMASSIVE
SPHALERITE**SPHALERITE**

Cubic • H 3½–4 • SG 3.9–4.1

A sulfide of zinc with variable iron content, sphalerite is the most heavily mined zinc-containing ore.

COMMON, MASSIVE
FORM OF
CHALCOPYRITE**CHALCOPYRITE**Tetragonal
H 3½–4 • SG 4.35

A sulfide of copper and iron, chalcopyrite has a deep brassy yellow colour. It has significant value as a copper ore.



CHALCOPYRITE CRYSTALS

ACANTHITEMonoclinic
H 2–2½ • SG 7.22

A sulfide occurring as dark, metallic, sometimes spiky crystals, acanthite is the main ore of silver.

